EXPERIMENT 1:

IMPLEMENTATION OF CAESAR CIPHER

AIM:

To implement the simple substitution technique named Caesar Cipher using C language.

ALGORITHM:

1. Start the program.
2. Read the plaintext message from the user.
3. Read the key value (shift value).
4. For each character in the plaintext:

    - If it is an uppercase letter, shift it by the key value.
    - If it is a lowercase letter, shift it by the key value.
    - Leave spaces and special characters unchanged.

5. Display the encrypted ciphertext.
6. Stop the program.

PROGRAM:

```
#include <stdio.h>

    #include <string.h>



    int main()

    {

        char text[100];

        int key, i;
```

```c
    printf("Enter the plaintext: ");

    fgets(text, sizeof(text), stdin);



    printf("Enter the key: ");

    scanf("%d", &key);



    for (i = 0; text[i] != '\0'; i++)

    {

        if (text[i] >= 'A' && text[i] <= 'Z')

            text[i] = (text[i] - 'A' + key) % 26 + 'A';



        else if (text[i] >= 'a' && text[i] <= 'z')

            text[i] = (text[i] - 'a' + key) % 26 + 'a';

    }



    printf("Encrypted text: %s", text);



    return 0;

}
```

OUTPUT:

Output Clear

```
Enter the plaintext: Kanmani
Enter the key: 6
Encrypted text: Qgtsgto


=== Code Execution Successful ===
```

RESULT:

Thus the implementation of Caesar cipher had been executed successfully.